cellqc QC report

Single-cell RNA-seq quality control

2026-08-05

1 sample | cellqc v0.3.0 | random seed 42

What this run did

Pipeline

1. Ambient RNA: **soupx**
   compared (not applied): decontx
2. Filter on counts, genes, % mitochondrial
3. Doublets: doubletfinder, scdblfinder
   cells removed by **doubletfinder** only
4. Nuclear fraction: 1/1 sample

Thresholds

- min UMI per cell: 500
- min genes per cell: 300
- max % mitochondrial: 10
- expected doublet rate: 0.1 at 13000 cells/reaction

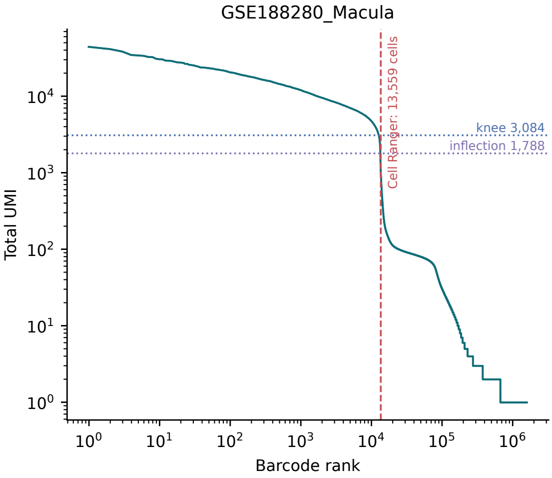
Cells retained at each stage

Sample	Cell Ranger cells	After count filter	Removed (count)	After doublet	Removed (doublet)	Retained
GSE188280_Macula	13559	11234	2325	10223	1011	0.754

Every exclusion is the difference between two adjacent columns.

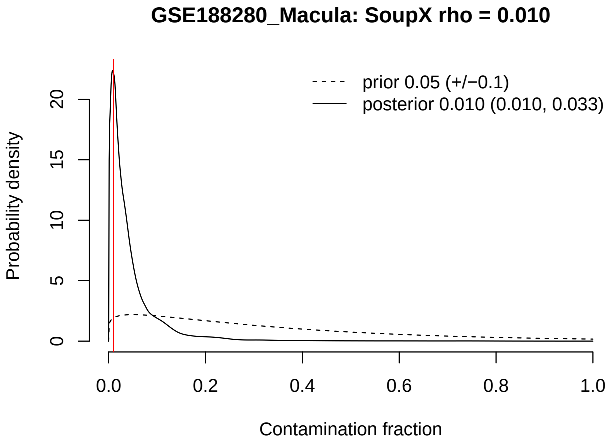
GSE188280_Macula: Cell Ranger summary

Metric	Value	Metric	Value
Estimated Number of Cells	13,559	Reads Mapped to Genome	96.7%
Mean Reads per Cell	17,203	Reads Mapped Confidently to Genome	93.3%
Median Genes per Cell	2,767	Reads Mapped Confidently to Intergenic Regions	6.3%
Number of Reads	233,256,198	Reads Mapped Confidently to Intronic Regions	53.2%
Valid Barcodes	97.3%	Reads Mapped Confidently to Exonic Regions	33.8%
Valid UMI Sequences	99.9%	Reads Mapped Confidently to Transcriptome	63.0%
Sequencing Saturation	28.1%	Reads Mapped Antisense to Gene	23.4%
Q30 Bases in Barcode	95.9%	Fraction Reads in Cells	90.6%
Q30 Bases in RNA Read	93.6%	Total Genes Detected	33,530
Q30 Bases in UMI	95.7%	Median UMI Counts per Cell	6,163



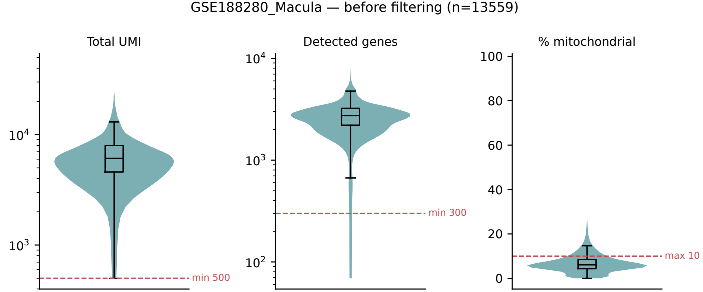
Cell Ranger EmptyDrops call (dashed line) against the UMI-rank curve. A clean library shows the call near the knee. Diagnostic only — cellqc does not re-call cells.

GSE188280_Macula: Ambient RNA (soupx)



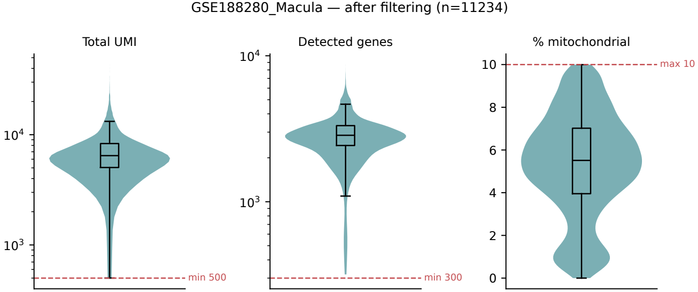
Estimated contamination fraction. This correction WAS applied to the counts.

GSE188280_Macula: QC metrics before filtering

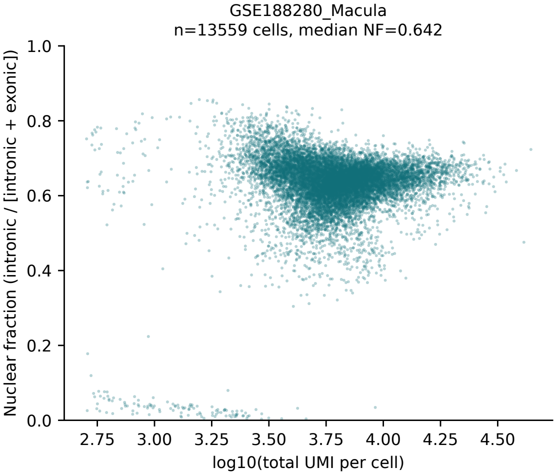


Dashed lines are the applied thresholds.
Counts and genes on a log scale.

GSE188280_Macula: QC metrics after filtering

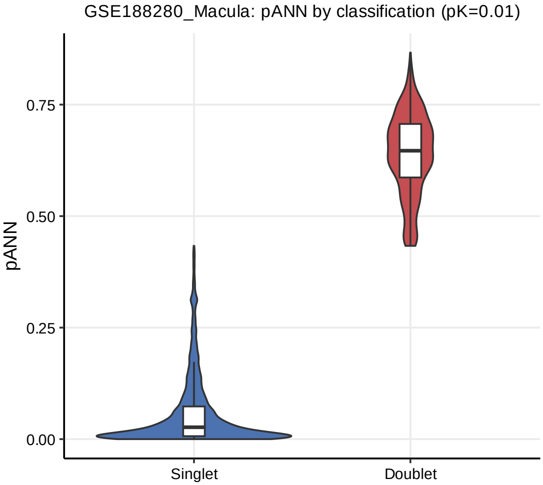


Same axes as the previous slide, after removing cells failing any threshold.



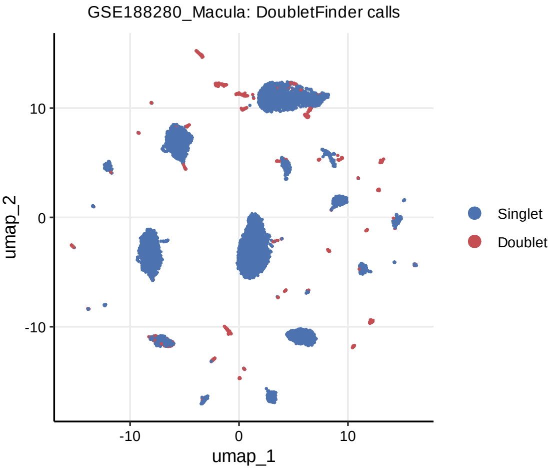
Intronic / (intronic + exonic) reads per cell against sequencing depth. High nuclear fraction at low depth suggests damaged cells or free nuclei. Reported only — NOT used for filtering.

GSE188280_Macula: DoubletFinder: pANN



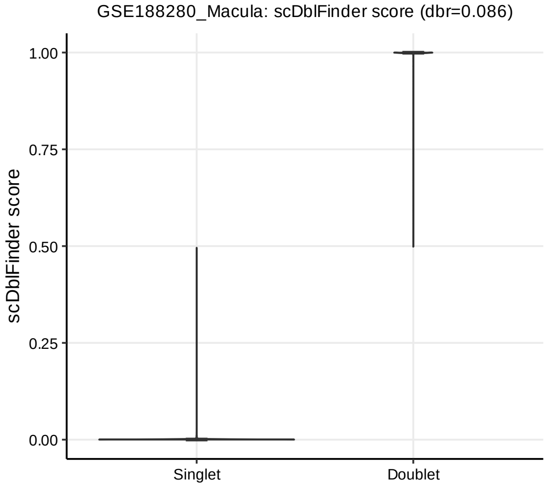
Proportion of artificial nearest neighbours, by call.

GSE188280_Macula: DoubletFinder: calls on UMAP



Doublets should sit between transcriptionally distinct populations, not form their own island.

GSE188280_Macula: scDblFinder: score



Run with the same expected doublet rate as DoubletFinder so the two are compared under matched assumptions. Diagnostic only.

Limitations to carry with these results

- Homotypic doublets are not modelled (modelHomotypic is deliberately not called), so the expected-doublet count over-estimates the DETECTABLE doublet count and the doublet step removes slightly more cells than the true heterotypic count. The bias direction is known and constant.
- The expected doublet rate is the 10x rule of thumb (rate=0.1, capacity=13000 cells per reaction), not a measurement for this library.
- Cell calling is Cell Ranger EmptyDrops. cellqc does not re-call cells; the barcode rank plot is diagnostic only.
- The nuclear fraction is reported but NOT used for filtering. DropletQC-style empty-drop and damaged-cell thresholds are sample- and tissue-dependent, so applying them automatically across a cohort would be unreviewed auto-filtering.
- Two doublet callers ran; only doubletfinder removed cells. The concordance table is a consistency measure, NOT evidence that either caller is correct – with no ground-truth doublets neither can be shown superior on this data.
- Ambient correction applied: soupx. Also estimated for comparison: decontx (these did NOT modify counts).
- Random seed 42 was set for every stochastic step. v0.1.0 set no seed and its doublet calls were not reproducible.